Sahaj Bhadja

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Summary

Full-stack engineer experienced in **TypeScript, React, React Native, and Next.js**, shipping customer-facing product features, PostgreSQL and Firebase-backed services, and LLM-powered workflows end-to-end.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Built Python FastAPI services and PostgreSQL schemas for a multi-tenant platform serving 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through query optimization end to end.
- Implemented cloud data pipelines processing 5,000+ monthly jobs with validation, reconciliation, and CI/CD-backed releases, increasing concurrent throughput 3x while achieving 99%+ reliability for recurring integrations at production scale.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Designed React interfaces and Python Flask backends for an AI grading product serving 150+ instructors, processing 10,000+ daily requests with MySQL persistence and stable API performance at scale.
- Integrated OpenAI LLM APIs and prompt chains for answer generation, rubric generation, grading, and feedback, adding documented fallbacks while reducing customer feedback turnaround time 60% at scale.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Built Jenkins CI/CD tooling and 20+ unit and integration tests for Java and Python microservices, increasing pipeline efficiency 68% while collaborating with Microsoft and GitHub stakeholders at scale.
- Optimized large-scale API integrations using Python, AWS Glue, and S3, improving response times 30% for low-bandwidth users while reducing technical debt 20% across production ETL workflows end to end.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a React and Node.js resource management platform with REST APIs and eight microservices, syncing two million records while reducing resource wastage 42% and increasing conversions 15% at scale.
- Shipped full-stack product features end to end using React, Docker, and Jenkins, cutting deployment time 50% while using analytics insights to prioritize product improvements in production environments.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Analyzed client requirements, mapped 12+ integration points, and implemented React Native and TypeScript React booking and payment experiences with Firebase Firestore sync for end users at scale.
- Delivered Python Flask and Firebase solutions across 50,000+ booking and payment records, validating implementation against user needs while achieving sub-200-millisecond responses and reducing manual refund handling 70% at production scale.

Skills

Languages: TypeScript, JavaScript, Python, SQL, Java
Frontend: React, React Native, Next.js, HTML, CSS
Backend: Node.js, Flask, FastAPI, REST APIs
Data: PostgreSQL, Firebase, Firestore, MongoDB, GCP
AI: OpenAI API, LLM APIs, NLP, LLM Systems
Delivery: CI/CD, Docker, Kubernetes, Git, Jenkins, Agile, AWS, Azure

Projects

FinNexus Live | *Next.js, TypeScript, Finance*
- Built a Next.js TypeScript fintech platform supporting 50,000+ users and one million transactions, implementing Plaid bank linking, MongoDB persistence, and CI/CD automation for reliable customer delivery at scale.

Big-Brain Live | *GCP, PostgreSQL, LLM Product*
- Built a customer-facing AI document platform on GCP with PostgreSQL, LLaMA-2, and FAISS, enabling conversational search and summarization for 200+ users while reducing document-analysis time 60% at scale.

PiXMod Live | *Next.js, Product*
- Designed a Next.js SaaS image platform with Stripe subscription billing and MongoDB storage, processing 1,000+ edited assets while shipping a responsive customer-facing interface for end users at scale.